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CNet-Cox for interpretable network biomarker discovery and survival risk scoring in precise breast cancer prognosis

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Lingyu Li^{1,2,3}, Weiqin Zhao⁴, Qingpeng Zhang^{5,6}, Wai-Ki Ching²✉ & Zhi-Ping Liu^{1,7}✉

Biomarker discovery in biomedicine is often cast as feature selection, yet most methods overlook gene co-localization within regulatory interaction networks, yielding isolated biomarkers with limited biological interpretability and clinical translatability. Here, we propose **CNet-Cox**, a disease-agnostic, **C**onnecte**d** **N**etwork-regularized **C**ox proportional hazards framework that incorporates prior network connectivity into sparse feature selection to identify connected prognostic module. Applied to breast cancer, CNet-Cox revealed the network structure of 68 prognostic biomarkers associated with survival on discovery dataset (TCGA, $n = 1092$) and achieved a concordance index of 0.913 on internal test dataset, outperforming conventional regularized Cox methods. From these network biomarkers, we derived a six-gene prognostic risk score (PRS) and validated its robustness across seven independent bulk transcriptomic datasets (GEO; $n = 2142$) and a spatial transcriptomics dataset (Visium; 4992 spots). The PRS consistently improved risk stratification (log-rank $p < 0.05$) and produced concordant predictions with MammaPrint in spatial prognostics (Pearson $r = 0.993$). Although evaluated in breast cancer, CNet-Cox is readily extensible to other diseases, molecular interaction networks and time-to-event endpoints, providing a generalizable tool for digital pathology and precision oncology. Overall, our comprehensive downstream analyses highlight that CNet-Cox offers a novel network-aware survival model for systematically discovering connected biomarkers and delivering scalable, precise and interpretable risk prediction.

Breast cancer (BRCA) is a multifactorial complex disease and the second leading cause of global cancer incidence in 2022, with approximately 2.3 million new cases, accounting for 11.6% of all cancer cases¹. Timely diagnosis and precise prognostic biomarkers are crucial for guiding treatment decisions and improving patient outcomes². However, the gold standard for BRCA diagnosis, typically involving mammography, ultrasound, and biopsy, can be invasive and costly, limiting its widespread clinical use³. Consequently, there is an urgent need for non-invasive detection methods with high sensitivity and specificity⁴. In recent years, various approaches have been developed to identify prognostic biomarkers for BRCA, including traditional histopathological evaluations and current genomic, proteomic, and artificial intelligence methods^{5–7}. Currently, the analysis of gene expression data increasingly emphasizes the relationship between gene expression and survival phenotypes, such as the time to BRCA recurrence or

death⁸, to identify novel biomarkers that can enhance BRCA prognosis and treatment.

Effective disease management through continuous prognostic monitoring is vital for achieving favorable clinical outcomes⁹. Currently, clinical prognosis prediction relies on clinicians' subjective assessments based on indicators, such as tumor size, TNM staging, and histopathological findings, which can be inaccurate¹⁰. Therefore, it is essential to develop more precise techniques for predicting patient outcomes and categorizing them into different risk levels to ensure appropriate treatment interventions¹¹. However, machine learning and feature selection algorithms are pivotal in identifying potential prognostic biomarkers from high-throughput datasets¹², thereby improving clinical decision-making, treatment selection, and prognosis for BRCA patients¹³. Through high-throughput screening methods, several prognostic biomarkers for BRCA have been

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¹Department of Biomedical Engineering, Shandong University, Jinan, Shandong, China. ²Department of Mathematics, The University of Hong Kong, Hong Kong SAR, China. ³School of Biomedical Sciences, The University of Hong Kong, Hong Kong SAR, China. ⁴School of Computing and Data Science, The University of Hong Kong, Hong Kong SAR, China. ⁵Musketeers Foundation Institute of Data Science, The University of Hong Kong, Hong Kong SAR, China. ⁶Department of Pharmacology and Pharmacy, The University of Hong Kong, Hong Kong SAR, China. ⁷National Center for Applied Mathematics, Shandong University, Jinan, Shandong, China. ✉e-mail: wching@hku.hk; zpliu@sdu.edu.cn

identified, with the estrogen receptor (ER) being one of the most common. Other widely used markers include the progesterone receptor (PR) and human epidermal growth factor receptor 2 (HER2)¹⁴. Nonetheless, relying solely on a single dataset or a specific type of information may not ensure the accuracy of the selected biomarkers¹⁵.

To address this limitation, integrating genomic data with prior knowledge is essential for accurately identifying BRCA biomarkers^{16–18}. For instance, mutations in genes like *BRCA1* and *BRCA2* are strongly associated with an increased BRCA risk and poor prognosis¹⁹. Tools such as MammaPrint²⁰ help identify genes linked to recurrence risk, while databases like KEGG²¹ provide insights into biological pathways involved in breast cancer¹⁵. This prior knowledge aids in uncovering critical genes associated with cancer dysfunction²², enabling the selection of reliable, biologically interpretable biomarkers²³. By leveraging these resources, researchers can enhance the accuracy and relevance of biomarker discovery.

However, traditional methods for identifying prognostic biomarkers often focus on individual genes related to disease phenotypes, overlooking the complex nature of gene interaction networks^{24,25}. Gene-gene interactions influence BRCA susceptibility and severity, affecting prognosis. Networks illustrating these interactions reveal how genes collaborate, highlighting systemic processes crucial for life^{26,27}. Incorporating these networks into feature selection improves prognostic biomarker identification, aiding in prediction and prevention strategies. Although the Cox Proportional Hazards (CPH) model is widely used for survival analysis (time-to-event data)²⁸, its traditional form lacks feature selection, often resulting in “independent” or “isolated” biomarkers²⁹. Integrating network structures into such models ensures better interpretability and elucidates molecular mechanisms, thereby enhancing BRCA prognostic research through connected network constraints.

In this study, bulk RNA-seq, microarray, and spatial transcriptomics (ST) datasets were collected and utilized to discover prognostic biomarkers from a cohort of 1093 participants (discovery dataset), 2142 participants (external validation dataset) and 4992 spots (ST validation dataset), accounting for right-censoring and censoring times. A Connetted Net work-regularized Cox proportional hazards (CNet-Cox) model, leveraging prior information and gene interaction networks, was developed through machine learning using an RNA-seq dataset from TCGA and rigorously validated across internal and external datasets. Notably, CNet-Cox outperformed conventional regularized Cox methods, including Lasso, Elastic net (ENet), L_0 , $L_{1/2}$, MCP, and SCAD, in accurately identifying BRCA patient survival states. Additionally, the CNet-Cox framework outperformed conventional single-gene approaches by leveraging network topology to identify a robust panel of prognostic biomarkers, resulting in a markedly higher concordance index for clinical outcome prediction. Furthermore, the derived prognostic risk scoring (PRS) system further facilitated nuanced and dependable patient stratification across diverse independent validation datasets, aligning closely with MammaPrint predictions in spatial tissue contexts with distinct cell types. Collectively, our downstream analyses, including tumor metastasis validation, protein profiling, meta-drugs screening analysis and external dataset validation, highlight CNet-Cox as a powerful and innovative tool for comprehensive biomarker discovery and precision prognostication in BRCA.

Results

High-level description of CNet-Cox and study design for breast cancer prognostic biomarker discovery

To demonstrate how CNet-Cox can be used as a general framework for discovering and validating prognostic network biomarkers from high-throughput data, we established an end-to-end analysis pipeline for breast cancer (Fig. 1A). The discovery set consisted of bulk RNA-seq data from 1,093 TCGA-BRCA patients, and the validation sets included seven independent GEO microarray cohorts (overall $n = 2,142$) and one 10x Visium spatial transcriptomics (ST) dataset (4992 spots). As a first step, gene expression data and associated clinical information were downloaded from the TCGA and GEO databases and preprocessed. Differential expression

analysis was then performed to identify differentially expressed genes (DEGs; Fig. 1B①). Next, these DEGs were combined with BRCA-related genes curated from KEGG²¹, MammaPrint²⁰, OSbrca³⁰, scPrognosis³¹, and transcription factors (TFs) from the updated RegNetwork database³². To strengthen biological plausibility and connectivity, the integrated gene set was mapped onto gene regulatory interactions from RegNetwork, yielding a network of 544 interconnected genes with matched expression profiles (Fig. 1B②). Subsequently, the CNet-Cox model was trained on the TCGA discovery cohort for network-regularized feature selection and survival analysis, and its prognostic performance was assessed by the concordance index (C-index; Fig. 1B③). Applying CNet-Cox to the 544-gene network identified 68 genes with non-zero regression coefficients that form a single connected prognostic module (Fig. 1B④; see also “CNet-Cox robustly identifies network-connected prognostic biomarkers for breast cancer and outperforms conventional regularized Cox models”). From these 68 network biomarkers, we further derived a parsimonious six-gene prognostic risk score (PRS) using univariate and multivariate Cox regression analyses (Fig. 1B⑤; see “CNet-Cox yields a six-gene prognostic risk score validated by nomogram and calibration for personalized breast cancer survival prediction”). The prognostic significance of the PRS genes was examined through nomogram prediction, analyses of breast cancer metastasis, and validation of protein expression profiles. Finally, the robustness and effectiveness of the PRS indicator for BRCA prognosis were evaluated across seven independent GEO cohorts and in the Visium ST dataset, where we assessed risk stratification performance and concordance with established signatures such as MammaPrint (Fig. 1B⑥; see “PRS score enables accurate breast cancer risk stratification and subtyping in multicohort and spatial transcriptomic validation”).

CNet-Cox robustly identifies network-connected prognostic biomarkers for breast cancer and outperforms conventional regularized Cox models

We initiated the exploration by analyzing a biomarker discovery dataset based on DEGs (Fig. 2A), priori BRCA-related genes, and gene interaction network to identify potential prognostic biomarkers from BRCA gene expression and clinical follow-up data (see Methods). The dataset was divided into training and testing sets in a 7:3 ratio to facilitate the training and evaluation of CNet-Cox model. In addition, the optimal tuning parameter for CNet-Cox was determined using five-fold cross-validation log partial likelihood (CV-LPL)³³ on the training dataset and validated on the testing dataset. The results showed that the CV-LPL value, as shown in Equation (14), reached its minimum when $\alpha = 0.500$ and $\lambda = 2.4273$. Ultimately, the number of features selected by the CNet-Cox model corresponds to those with substantial coefficients θ_j , specifically, the value of regression coefficients θ_j satisfying $|\theta_j|/\|\theta\|_\infty \geq \tau$ for $j = 1, 2, \dots, 544$, with coefficients not meeting this criterion set to zero³⁴. Here, we chose $\tau = 5.15 \times 10^{-2}$ because, through empirical testing, it yields optimal model performance by balancing the inclusion of meaningful features and avoiding overfitting (see Methods), effectively filtering out features with negligible influence to retain only those with substantial contributions for downstream analysis.

The CNet-Cox method identified 68 genes as prognostic biomarkers for BRCA, with their network structure clearly depicted (Fig. 2B), where node size is proportional to its degree and colors indicate different sources of information. These include 32 TF genes from RegNetwork (red), 27 genes from KEGG (green), 10 from OSbrca (purple), and 5 from MammaPrint (range). Notably, genes like *EGFR* and *NCOA3* appear in both OSbrca and KEGG, while *JUN*, *FOS*, *SP1*, and *MYC* are shared between TFs and KEGG, emphasizing their relevance to BRCA. It is crucial to recognize that uninformed feature selection methods typically fail to identify TFs, likely because of the low signal-to-noise ratio stemming from their generally low expression. Thus, the inclusion of these TFs underscores the benefits of our integrative approach in detecting biologically significant genes that conventional methods might miss. Besides, the identified biomarkers do not overlap with those from differential expression analysis shown in Fig. 2A, highlighting the importance of integrating prior disease knowledge and gene

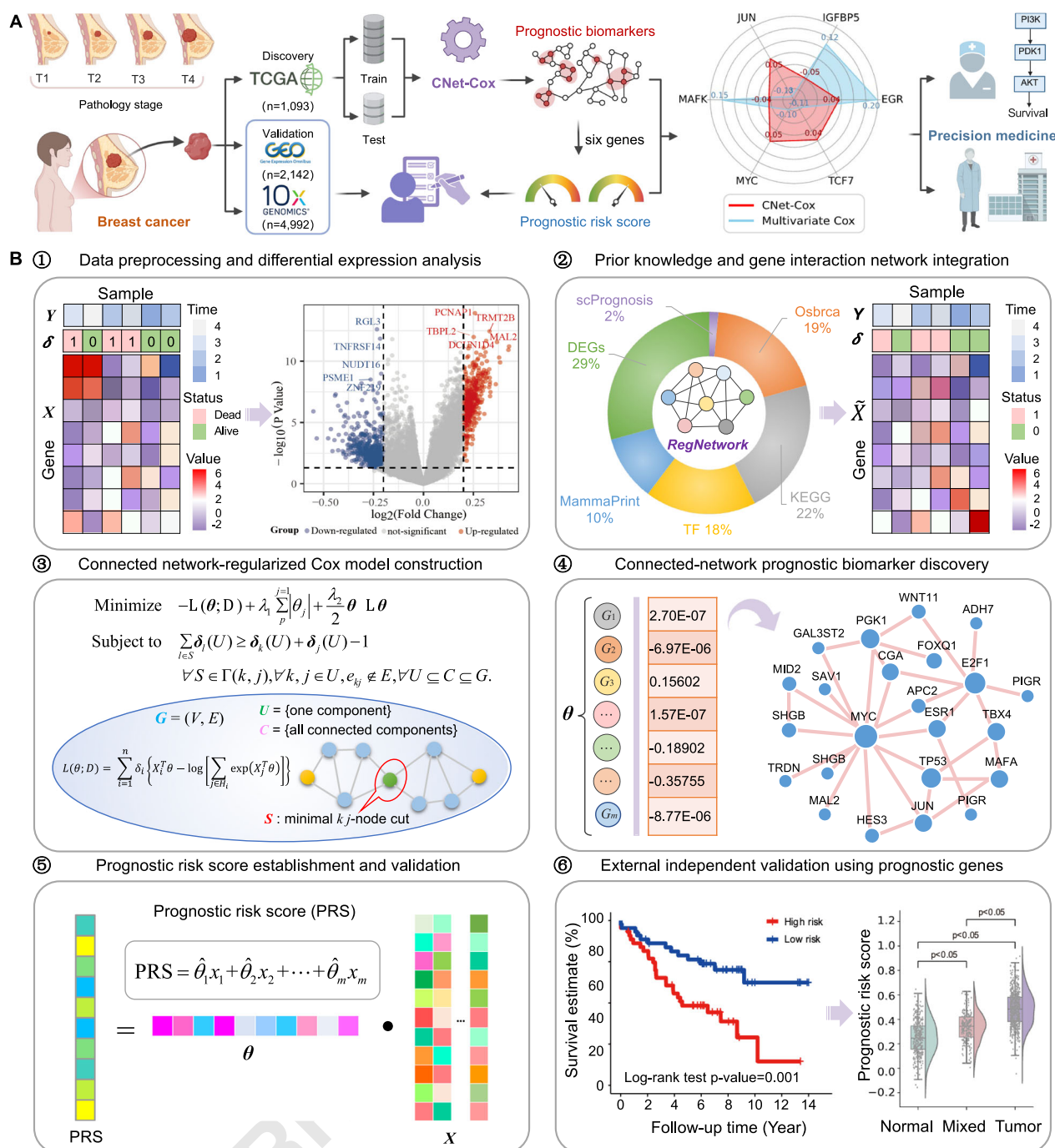


Fig. 1 | Study design and workflow of identifying and verifying prognostic biomarkers from CNet-Cox. **A** Study design. Patients or samples are collected from three databases, and prognostic biomarkers are identified using CNet-Cox and validated with internal dataset. Subsequently, PRS system was established for risk estimation, followed by independent external validation. Part of this panel is created in BioRender. Huang, Y. (2026) <https://BioRender.com/cv9kqhqt>. **B** Workflow of CNet-Cox for network-aware prognostic biomarker discovery and risk scoring. ① Identify differentially expressed (up-/down-regulated) genes by comparing gene

expression between deceased and surviving patients at the last follow-up. ② Integrate prior knowledge and gene interactions to build a network. ③ Construct CNet-Cox with sparsity and graph penalties under connectivity constraints. ④ Selected connected prognostic genes are mapped onto the network. ⑤ Compute PRS as a weighted sum of selected expressions from Cox models. ⑥ Validate externally, showing clear high- vs. low-risk separation and PRS differences across normal, mixed, and tumor samples.

interaction networks to select reliable biomarkers. The network's degree distribution follows a power-law (Fig. 2C), and betweenness centrality is strongly correlated with node degree ($R = 0.8466$; Fig. 2D). These properties are characteristic of scale-free biomolecular networks, suggesting that the 68-gene prognostic module identified by CNet-Cox preserves typical regulatory-network topology rather than representing a random set of genes. In particular, the presence of hub TFs and BRCA-related genes at the

core of the module supports its biological plausibility and indicates that CNet-Cox recovers non-random, topologically coherent network structures (see Methods).

To emphasize the key findings from comparing the CNet-Cox model with other regularized Cox models in feature selection, a visualization of the structure within selected features was performed (Fig. 2E, Supplementary Table 1). Numerically, Fig. 2F displays the results across eight regularized

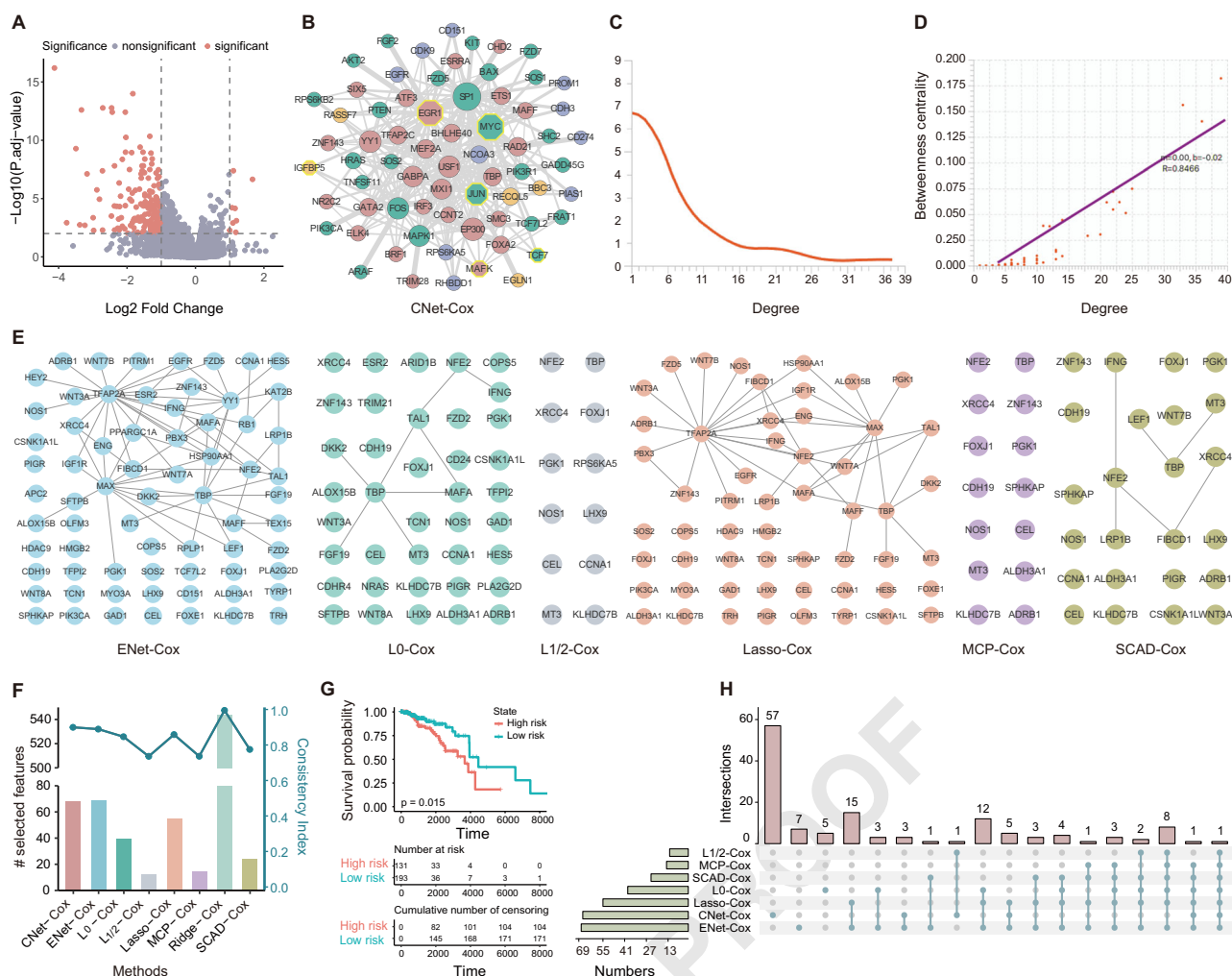


Fig. 2 | CNet-Cox identified 68 feature genes with network structure as prognostic biomarkers of breast cancer. **A** Differential expression analysis. **B** Network structure of 68 features identified using our proposed CNet-Cox method. **C** Degree distribution curve of network nodes. **D** Betweenness centrality of characteristic

network. **E** Network structure of selected features using the other six regularized Cox methods with different penalties. **F** Comparison results of feature selection using different methods. **G** Survival analysis results of 68 prognostic BRCA markers. **H** Overlap of features identified by different regularized Cox methods.

Cox methods, with the number of selected feature genes shown on the left vertical axis and the C-index on the right. For comparison, the time-dependent area under the ROC curve at 1-, 3-, and 5-year time points for all methods is shown in Supplementary Fig. 1. Notably, the $L_{1/2}$ -Cox method selects the fewest features, while Ridge-Cox method includes the most, due to its approach of maintaining coefficient smoothness without shrinking them to zero. Importantly, the CNet-Cox method, despite selecting one fewer feature than the ENet-Cox method, achieves a superior C-index, exceeding 0.900 and outperforming ENet-Cox by about one percentage point. Although Ridge-Cox attains the highest C-index, the CNet-Cox follows closely, ranking as a top performer with a C-index of approximately 0.913. In contrast, the $L_{1/2}$ -Cox and MCP-Cox methods perform less effectively, with C-indices below 0.800, highlighting the superior prognostic capability of the CNet-Cox approach (Supplementary Table 2, Supplementary Fig. 2). Fig. 2

G presents the survival curves derived from the prognostic biomarkers identified by the CNet-Cox model within the discovery dataset. The Kaplan-Meier (KM) analysis clearly separates BRCA samples into high-risk and low-risk groups, with a statistically significant Log-rank test P value of approximately 0.015. Additionally, Fig. 2H illustrates the overlap among CNet-Cox and six other regularized Cox feature selection methods, emphasizing interactions with the top bars and dashed lines below. Significantly, all seven methods consistently identify the gene *TBP*, known to

compete with the BRCA prognostic biomarker *MBP-1*, a transcriptional repressor with antiproliferative effects on breast cancer cells³⁵. Furthermore, the intersection between Lasso-Cox and ENet-Cox methods is the largest, sharing 15 overlap genes. The CNet-Cox model stands out by sharing five genes with Lasso-Cox and ENet-Cox while uniquely identifying 57 additional genes, demonstrating its robust feature selection capability.

CNet-Cox reveals functionally enriched and semantically similar prognostic biomarkers associated with anthracycline resistance in breast cancer

To investigate the pathological significance of the 68 BRCA prognostic biomarker genes identified by the CNet-Cox method, pathway and process enrichment analyses were performed using Metascape (see Methods)³⁶. Figure 3A displays a heatmap of the top 20 enriched clusters, where the color depth corresponds to the P value from the cumulative hypergeometric test. It is evident that the most significant pathway enriched is “hsa05224: Breast cancer” from KEGG Pathway ($P \approx 1.0e-44$) (Supplementary Fig. 3). The CNet-Cox method, though similar to the existing ENet-Cox method in terms of the number of identified features and C-index, exhibits distinct advantages in functional enrichment analysis of genes. Specifically, CNet-Cox demonstrates more significant enrichment results for key pathways, such as the “hsa05224: Breast cancer” pathway from KEGG, compared to ENet-Cox ($P = 2.5e-06$, Student’s t -test, Fig. 3B). Importantly, CNet-Cox

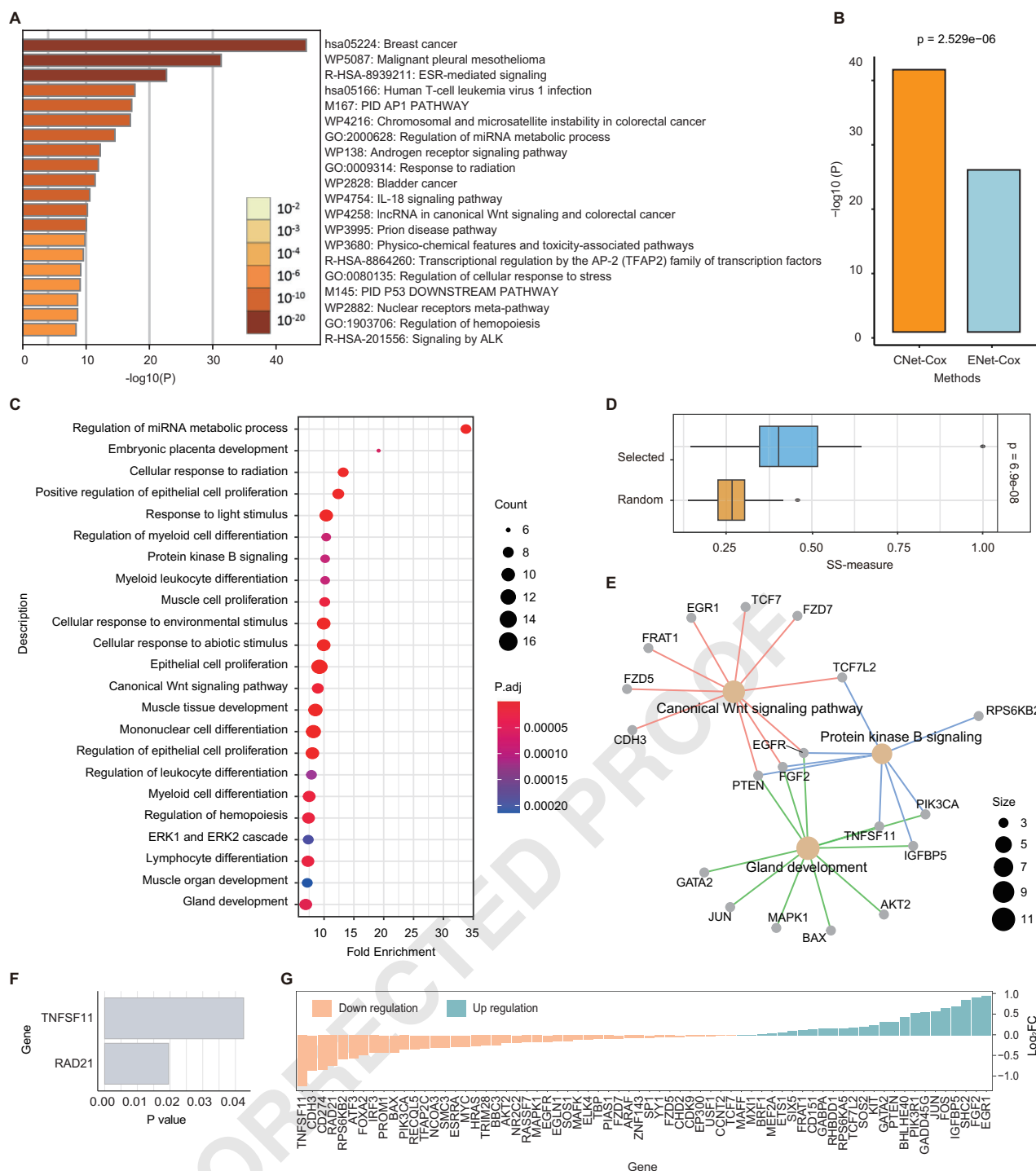


Fig. 3 | Biological process enrichment and drug sensitivity/resistance analysis for 68 prognostic biomarker genes. A Heatmap of the top-enriched 20 terms. **B** Significance comparison of enrichment analysis (CNet-Cox vs ENet-Cox). **C** Top 23 terms of GO biological process enrichment analysis. **D** Semantic similarity results

of 23 terms. **E** Gene network of three GO terms with semantic similarity greater than 0.500. **F** Two significantly differentially expressed genes. **G** Analysis of up- and down-regulation of 68 prognostic markers between anthracycline-sensitive and anthracycline-resistant groups.

and ENet-Cox are both applied to the identical input dataset enriched with cancer-relevant genes from prior biological knowledge. The main difference is that CNet-Cox integrates gene connectivity information from gene-gene interaction network during feature selection, while ENet-Cox considers genes as independent features, ignoring their relationships or network structure. This network constraint allows CNet-Cox to pinpoint functionally coherent and biologically interpretable biomarker modules, leading to improved enrichment and offering deeper insights into the biological importance of prognostic biomarkers in BRCA.

To further clarify the role of prognostic biomarkers within BRCA signaling pathways, we analyzed the top five enriched pathways for their connections to breast cancer. These pathways, sourced from WikiPathways, Reactome Gene Sets, KEGG Pathway, and Canonical Pathways, are each supported by previous studies linking them to breast cancer pathology and progression^{37–40} (as detailed in the Methods). Their intricate association with BRCA is emphasized by the functional enrichment heatmap (Supplementary Fig. 4A), where each node represents an enriched Gene Ontology (GO) term, with node size proportional to the number of associated genes. To

further explore the biological functions of the 68 biomarkers, we conducted a protein-protein interaction (PPI) enrichment analysis using the STRING, BioGrid, OmniPath, and InWeb_IM databases³⁶, resulting in a network comprising 63 genes and 255 edges (Supplementary Fig. 4B). Subsequently, the MCODE algorithm identified four distinct connected components within this network (Supplementary Fig. 4C). Lastly, pathway and process enrichment analyses of these components (Supplementary Fig. 4D) revealed significant alignment with the pathways identified in the overall network. This alignment of 150 underscores the importance of the CNet-Cox method in enhancing the analysis of breast cancer biomarkers.

Having established the network topology and pathway associations of these biomarkers, we next sought to elucidate their functional implications at the biological process level. To investigate the potential pathological significance of the identified prognostic biomarker genes in BRCA, we conducted a GO functional enrichment analysis. Initially, this analysis was performed on the 68 genes, revealing 23 significantly enriched GO biological processes (Fig. 3C). The results indicate that breast cancer dysfunctions, based on prior knowledge, are enriched within the identified biomarkers, thus biologically validating their accuracy. This finding further demonstrates the efficacy of the proposed CNet-Cox method for prognostic biomarker discovery.

Furthermore, using Equation (3) in the Methods of Supplemental Materials, we calculated the semantic similarity measure⁴¹ (SS-measure) between the GO terms identified from the prognostic biomarker functional enrichment analysis and the 23 hallmark breast cancer GO terms documented in the literature^{42,43}, as listed in Supplementary Table 3. For comparison, we randomly sampled 23 terms from all cancer-related GO terms and computed their SS-measure with the specific hallmark terms. This sampling process was repeated 10 times to ensure general applicability, with the average score serving as the final SS-measure (see Methods). Figure 3D illustrates that the SS-measure of GO terms enriched by the identified BRCA prognostic biomarkers is significantly higher than that of randomly sampled terms ($P=6.9\text{e-}08$, Student's t -test), indicating a strong association with BRCA biological functions. Figure 3E presents a network of three GO Biological Process (BP) terms with SS-measures exceeding 0.600, along with their associated genes, highlighting their close relationship to BRCA onset and progression. This network demonstrates that genes are interconnected, collectively functioning through interactions, thereby providing insights into the molecular mechanisms underlying breast cancer development.

These findings not only elucidate the functional role of the identified biomarkers in breast cancer biology but also demonstrate their clinical relevance in addressing anthracycline resistance, a major therapeutic challenge in BRCA treatment. While anthracyclines remain first-line therapeutic agents due to their established antitumor efficacy, their clinical utility is significantly limited by the development of resistance. In a comprehensive study by Liu et al., relative expression matrices from the TCGA database were analyzed for 86 BRCA patients who showed a complete response to anthracycline therapy and 7 patients who developed resistance⁴⁴, which involved a comparative analysis of the divergent transcriptomic profiles between anthracycline-sensitive (anthracycline-S) and anthracycline-resistant (anthracycline-R) patients. Building on this work, we investigated the differences between anthracycline-S and anthracycline-R groups within the 68 genes identified by the CNet-Cox method and found that the P -values for DEGs calculated using DESeq2 indicated that *RAD21* and *TNFSF11* exhibit statistical significance (Fig. 3F; Supplementary Table 4). This finding suggests that these genes, identified by the CNet-Cox method, could serve as potential diagnostic and prognostic biomarkers for BRCA patients with anthracycline resistance, potentially contributing to resistance mechanisms and limiting treatment success. Additionally, Fig. 3G provides statistics on the overexpression and underexpression of the 68 genes, showing that among these biomarkers, 25 genes are up-regulated and 43 are down-regulated, including the downregulation of *RAD21* and *TNFSF11*. Notably, these two genes were not identified by any of the six comparative regularized Cox methods other than Ridge-Cox, further

emphasizing the efficacy and superiority of the CNet-Cox method in identifying prognostic biomarkers for BRCA.

CNet-Cox yields a six-gene prognostic risk score validated by nomogram and calibration for personalized breast cancer survival prediction

The risk score based on the gene expression patterns of several biomarker genes plays a pivotal role in predicting cancer recurrence risk⁴⁵. Here, in order to develop a BRCA prognostic risk scoring (PRS) system, we conducted an analysis using 68 genes from 1,080 samples with clinical information in the TCGA database. Both univariate and multivariate Cox regression analyses were performed to assess overall survival (OS) and select key prognostic genes for constructing the PRS model. The univariate Cox regression analysis revealed that 39 genes with $P < 0.05$ were significantly associated with OS, with 15 genes positively correlated and 24 negatively correlated. Subsequently, multivariate Cox regression identified six independent prognostic genes (*EGR1*, *IGFBP5*, *JUN*, *MAFK*, *MYC*, and *TCF7*) that were highly correlated with OS ($P < 0.05$). The multivariate analysis results indicated that *EGR1*, *IGFBP5*, and *MAFK* are positively correlated with OS, while *JUN*, *MYC*, and *TCF7* are negatively correlated (Fig. 4A). Detailed results are available in Supplementary Table 5. Based on this analysis, we propose a PRS metric – a six-gene model:

$$\text{PRS} = 0.197 * x_{\text{EGR1}} + 0.120 * x_{\text{IGFBP5}} - 0.130 * x_{\text{JUN}} + 0.155 * x_{\text{MAFK}} - 0.099 * x_{\text{MYC}} - 0.109 * x_{\text{TCF7}}, \quad (1)$$

where x_{Gene} represents the expression value of the corresponding gene. Based on the PRS metric in Equation (1), all BRCA patients can be classified into high-risk or low-risk groups according to the optimal cutoff value of PRS, determined using X-Tile software^{46,47} (Supplementary Fig. 6).

To validate the effectiveness of the established PRS, we performed internal validation using discovery data. First, we extracted the expression values of the six genes in Equation (1) from the TCGA BRCA dataset and calculated PRS values using Equation (1). Patients were then categorized into high-risk and low-risk groups based on the selected threshold. Finally, the Kaplan-Meier curve (KM-curve) and significance of the survival analysis show a significant difference in survival probability between the two groups ($P = 0.006$), demonstrating the PRS index's efficacy in BRCA prognosis (Fig. 4B).

Then, we utilized nomograms to enhance clinical decision-making for BRCA treatment by incorporating individual patient characteristics, thereby improving survival rates and quality of life. We constructed grouping variables from clinical data in the TCGA database to assess the impact of the PRS metric on BRCA patient stratification (see Methods), where gender and pathologic stages were categorized into groups, facilitating the analysis of clinical features on prognosis through nomograms and calibration curves. Our nomogram, developed using the discovery dataset, assigns scores to various factors, with age emerging as the most significant predictor of outcomes (Fig. 4C), reflecting established BRCA risk patterns. This result aligns with documented facts: the risk of breast cancer increases with age, and most breast cancer diagnoses occur after the age of 50⁴⁸. Women who undergo natural menopause after age 55 have twice the likelihood of developing breast cancer compared to those who experience menopause before age 45⁴⁹. Pathologic M stage (indicating cancer metastasis) ranks second, and the PRS metric ranks third, with the length of the line segment for the PRS index slightly shorter than that for the pathologic M stage, indicating that the PRS index has considerable reference value in estimating survival probability.

To evaluate the accuracy of the PRS model, calibration curves were plotted, comparing predicted and actual survival probabilities for BRCA patients at 1, 3, and 5 years (Fig. 4D). The curves demonstrated strong alignment with actual outcomes, indicating high nomogram accuracy. Specifically, the calibration curves, with predicted probabilities on the x -axis and actual probabilities on the y -axis, showed that the red fitted line closely approximated the ideal 45-degree blue diagonal, confirming the model's

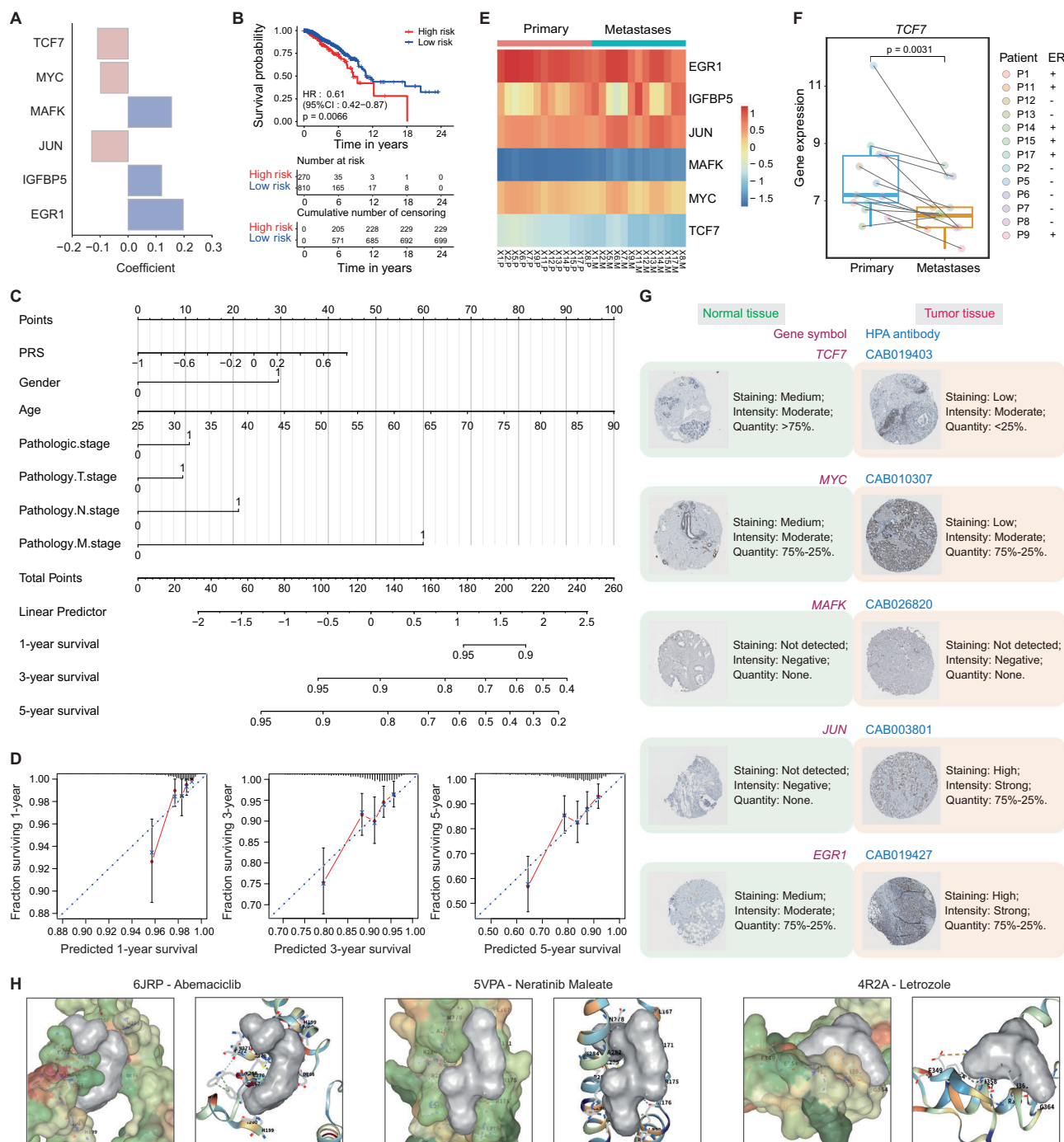


Fig. 4 | Comprehensive analyses of identified prognostic biomarkers, including PRS construction, internal and external validation, metastatic expression differences, immunohistochemistry distinction, and meta-drugs evaluation.

A Multivariate Cox analysis identifies six independent prognostic genes. **B** Internal validation result using six-gene PRS system. **C** The nomogram demonstrates that PRS plays a significant role in survival prediction. **D** Calibration curves for 1-, 3- and 5- year predictions indicate high accuracy of the nomogram. **E**, **F** Heatmap and

expression analysis in primary and metastatic patients reveal that *TCF7* is down-regulated in metastatic tumors. **G** Immunohistochemistry images sourced from HPA database (<https://www.proteinatlas.org/>) for prognostic genes show that *TCF7* exhibits lighter staining in tumor samples. **H** Three different molecular docking configurations with their interactions with target proteins from PDB indicate that Abemaciclib has the highest binding affinity for *TCF7*. Binding affinity image sourced from CB-Dock2 (<http://183.56.231.194:8001/cb-dock2/index.php>).

predictive reliability. These findings support the clinical utility of nomograms in personalizing BRCA treatment strategies and improving patient prognosis.

PRS gene orchestrates breast cancer progression via metastatic programming, clinical prognosis and actionable therapeutic targets

Cancer metastasis is a leading cause of cancer-related mortality, driven by certain transcriptional regulators that offer potential targets for treatment.

Zhang et al. analyzed 13 pairs of primary and metastatic BRCA tumor patients, among whom 6 have ER-positive (ER+) tumors and 7 have ER-negative (ER-) tumors, using RNA-seq to identify dysregulated genes and explore epigenetic drivers of metastasis⁵⁰. Their findings, supported by nomogram analysis, highlight the metastatic stage (M stage) as a significant factor in patient prognosis. Utilizing RNA-seq data (GSE148005) from the GEO database, we examined the relationship between six genes in the PRS metric and BRCA metastasis (Fig. 4E). Notably, the gene *TCF7* was

significantly downregulated in metastatic tumors compared to primary tumors ($P = 0.0031$, Student's t -test, Fig. 4F). An exception was observed in patient P13 (ER-), where gene expression was higher in the metastatic tumor sample than in the primary tumor sample. In contrast, for the other 12 patients, expression levels were decreased in metastatic tumor samples. This observation is consistent with findings in the literature⁵⁰, which report a down-regulation of immune-related pathways in metastatic breast cancer samples. Figure 4E presents the expression heatmap of the six genes in the PRS metric across BRCA tumor samples, highlighting their role in tolerance mechanisms. Among the PRS metric genes, *MAFK* is uniquely upregulated in metastatic tumors, aligning with Okita et al.'s research⁵¹ where quantitative polymerase chain reaction (qPCR) analysis revealed elevated *MAFK* expression in triple-negative Basal B-type breast cancer cell lines (Supplementary Fig. 5A). Moreover, Supplementary Fig. 5B demonstrates the knockdown of *MAFK* in 4T1 cells using two independent short hairpin RNAs (shRNA). After subcutaneously transplanting these cells into Balb/c mice, it was found that *MAFK* knockdown significantly reduced tumor volume (Supplementary Fig. 5C) and the number of metastatic nodules on the lung surface (Supplementary Fig. 5D). These findings underscore the potential for targeting *MAFK* in therapeutic strategies for metastatic breast cancer, with statistical significance confirmed by the values P calculated using Student's t -test.

The Human Protein Atlas (HPA) database categorizes immunohistochemistry results for proteins into four levels: not detected, low expression, medium expression, and high expression⁵². To validate the relationship between the six genes involved in Equation (1) and BRCA prognosis, we evaluated their staining patterns in breast cancer tumor tissues using the HPA database. Notably, the HPA lacks staining information for *IGFBP5* in breast tumors. However, previous research by Ghossaini et al. indicates that the increased BRCA risk associated with the g-allele of rs4442975 is mediated through reduced *IGFBP5* expression⁵³. Figure 4G illustrates the immunohistochemical staining patterns of the remaining five genes in both "normal" and "tumor" breast tissues, with images sourced from the HPA database (<https://www.proteinatlas.org/>). Figure 4G reveals that *TCF7* and *MYC* exhibit lighter staining in "tumor" samples, while *JUN* and *EGR1* show darker staining, consistent with their known association with BRCA^{54,55}. Although *MAFK* does not show significant expression differences between "normal" and "tumor" tissues, studies by Haakensen et al. suggest that miR-342-5p, which targets *MAFK*, is significantly linked to breast cancer progression in the Luminal B subtype⁵⁶. These findings propose *MAFK* as a potential prognostic biomarker for BRCA, though further research is necessary to confirm its role given that miR-342-5p targets multiple genes⁵⁶.

In order to advance targeted therapies, we employ molecular docking techniques to identify and evaluate metadugs, which are compounds designed to target multiple biological pathways or disease mechanisms concurrently. The CNet-RLR method significantly enhances drug discovery by efficiently narrowing down a vast array of compounds to those with substantial therapeutic potential. We found that the CNet-RLR method achieves this by examining interactions between drug candidates and target proteins. Specifically, taking the identified six prognostic genes as target proteins, we acquired 3D structures for *TCF7*, *MYC*, *JUN* and *EGR1* from the Protein Data Bank (using their respective PDB IDs: 6JRP, 6G6J, 5VPA and 4R2A), and for *MAFK* and *IGFBP5* from the AlphaFold Protein Structure Database (using their respective UniProt IDs: O60675 and P24593). Utilizing CB-Dock2⁵⁷, we performed molecular docking studies to examine the structural interactions between six receptors and 91 drug compounds approved by the Food and Drug Administration (FDA) for breast cancer treatment, where the binding affinities of the receptor-drug interactions are calculated (Fig. 4H, Supplementary Table 6). Among the compounds tested, Abemaciclib, Neratinib Maleate, and Letrozole demonstrated average docking scores ranging from -7.8 to -6.1 kcal/mol when interacting with the receptors *TCF7*, *JUN*, and *EGR1*. Notably, Abemaciclib demonstrated the highest affinity for *TCF7* at -7.8 kcal/mol. Based on these findings, we propose these three drugs as promising candidates for further investigation in breast cancer treatment.

PRS score enables accurate breast cancer risk stratification and subtyping in multicohort and spatial transcriptomic validation

To validate the effectiveness of the PRS score, a comprehensive analysis was conducted using six independent datasets from the GEO database, totaling 2142 samples. Similar to the internal validation process, the gene expression values for *EGR1*, *IGFBP5*, *JUN*, *MAFK*, *MYC*, and *TCF7* were extracted from the six independent datasets (Supplementary Table 7), and the PRS score was then calculated for each sample. Patients were divided into high-risk and low-risk groups based on an optimal PRS threshold, with significant differences in survival outcomes confirmed across datasets using the log-rank test ($P < 0.05$, Log-rank test). Detailed results, including KM-curves, Hazard Ratios (HR), 95% Confidence Intervals (CI), and P values, are shown in Fig. 5A, underscoring the PRS score's prognostic value for BRCA. Figure 5A also captures various survival metrics: Overall Survival (OS), Disease-Free Survival (DFS), Recurrence-Free Survival (RFS), and Distant Metastasis-Free Survival (DMFS), reflecting the multifaceted survival endpoints in BRCA patients (see Methods). Overall, the analysis reveals that the PRS metric has a significant impact on BRCA prognosis, demonstrating its potential utility in guiding treatment strategies and improving patient outcomes.

Evaluating the prognostic performance of our six-gene PRS index across major BRCA subtypes is crucial, given the biological heterogeneity of BRCA. Using 340 clinically annotated samples from the GSE96058 dataset with GPL18573 platform⁵⁸, we evaluated PRS risk stratification within estrogen receptor-positive (ER-positive) and triple-negative breast cancer (TNBC) patients (see Methods). Subtype classification was based on estrogen receptor (ER), human epidermal growth factor receptor 2 (HER2), and progesterone receptor (PR) status, whose proportions of 'Negative', 'Positive', and 'Missing' for each subtype are summarized in Fig. 5B. Our results show that PRS retains prognostic utility in both ER-positive and TNBC subtypes, with some performance differences. As shown in Fig. 5C, among ER-positive patients, PRS effectively stratifies individuals into high- and low-risk groups (log-rank $p = 0.015$). Similarly, in TNBC patients (Fig. 5D), PRS distinguishes risk groups with statistical significance (log-rank $p = 0.021$), though effect size and sample numbers vary.

Additionally, we also evaluated the PRS score using a spatial transcriptomics (ST) dataset from 10x Visium¹⁴. Guided by pathology annotations on H&E images, we selected three regions of interest (ROIs) representing distinct tumor subtypes: DCIS 1, DCIS 2, and invasive tumor cells (Fig. 6A), where the ductal carcinoma in situ (DCIS) is a non-obligate precursor to invasive ductal carcinoma, which can progress to invasive disease, typically treated with surgery and radiotherapy⁵⁹. Since not all DCIS lesions become invasive, understanding the molecular mechanisms driving invasiveness in DCIS is of great interest, though not well understood⁶⁰. Here, we aim to explore the PRS score's utility in identifying different tumor subtypes, thereby supplementing H&E imaging and pathology with molecular insights using ST data. In particular, we extracted spatial gene expression data for each ROI (Fig. 6B). Pathologists confirmed that these ROIs were morphologically distinct or had unique microenvironments, with DCIS 2 showing greater invasiveness than DCIS 1, and the invasive ROI displaying the highest incidence of invasive cell types^{14,61}. Notably, the spatial plot of six genes involved in the PRS score shows that *JUN*, *MYC*, and *TCF7* are inversely correlated with these ROIs (Supplementary Fig. 7A), consistent with their negative coefficients in the PRS score (see Fig. 4A). Interestingly, the PRS score is highest in the Invasive ROI compared to the two DCIS ROIs and is higher in DCIS 2 than DCIS 1, aligning with pathologists' annotations from histology images (Fig. 6C). Furthermore, the PRS score is inversely correlated with *SCGB2A2* and *CPB1* expression (marker genes for DCIS 1 and DCIS 2, respectively) and positively correlated with *CDH2* (marker gene for invasive tumor cells) (Supplementary Fig. 7B).

Especially, we evaluated spot-level scores for each cell type from the ST dataset using our PRS, as well as the established Oncotype DX 21-gene⁶² and MammaPrint 70-gene²⁰ signature scores (Supplementary Table 8, Supplementary Table 12). Notably, PRS and MammaPrint scores demonstrated

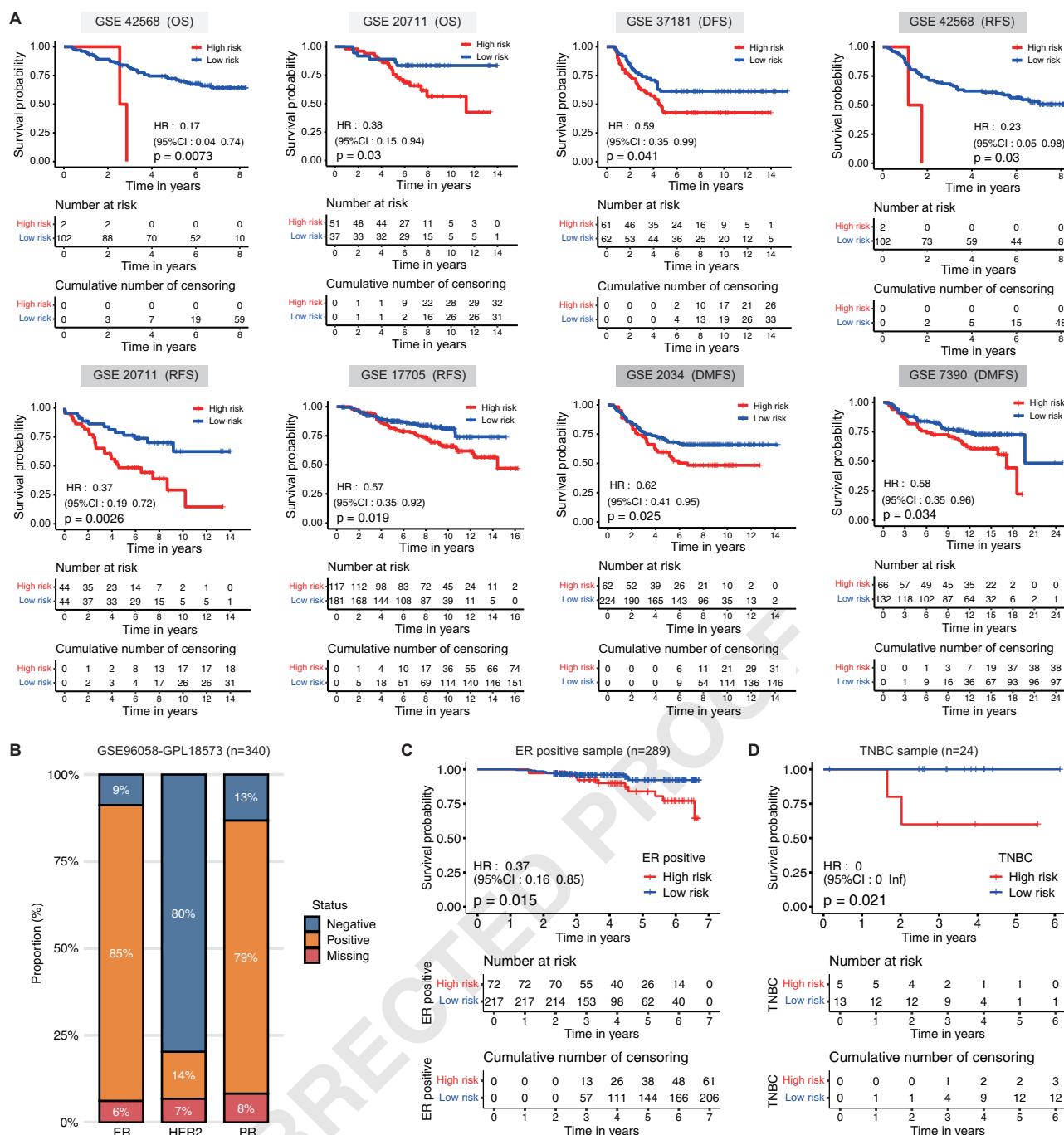


Fig. 5 | External independent datasets validate the prognostic effectiveness of the PRS in overall and major BRCA subtypes, specifically ER-positive and TNBC patients. A KM curves of high-risk and low-risk BRCA patients in external datasets (for OS, DFS, RFS, DMFS; Methods). **B** Stacked bar plot of 340 samples from the

GSE96058 dataset using the GPL18573 platform⁵⁸. **C, D** KM-curves of high-risk and low-risk groups among ER-positive **C** and triple-negative breast cancer (TNBC; **D**) BRCA patients in external datasets.

highly concordant prognostic assessments (Pearson correlation coefficient=0.993), whereas the Oncotype DX score failed to demonstrate invasive cells more invasive than DCIS cells (Pearson correlation coefficient=-0.774, Fig. 6D–F, where *** for $P < 0.001$; ** for $P < 0.01$; * for $P < 0.05$). To further explore the relationship among prognostic signatures, we analyzed gene overlap between our initial CNet-Cox biomarkers and those of Oncotype DX⁶² and MammaPrint²⁰. CNet-Cox shared five genes (*BBC3*, *EGLN1*, *IGFBP5*, *RASSF7*, *RECQL5*) with MammaPrint, but none with Oncotype DX (Fig. 6G). Among the six PRS genes, only *IGFBP5* overlapped with MammaPrint, while *EGFR1*, *JUN*, *MAFK*, *MYC*, and *TCF7* were unique to our model (Fig. 6H), suggesting their potential as novel prognostic

biomarkers for breast cancer. Overall, these results demonstrate the strong potential of the PRS for distinguishing both molecular tumor subtypes and clinically relevant BRCA subgroups, such as ER-positive and triple-negative breast cancers, thereby enhancing the precision of molecular profiling in breast cancer pathology.

Discussion

Breast cancer (BRCA) prognostic biomarkers are crucial for understanding patient incidence rates and predicting postoperative recurrence and survival periods. Traditional regularized Cox proportional hazards models often fail to consider gene-gene interactions when identifying cancer prognostic

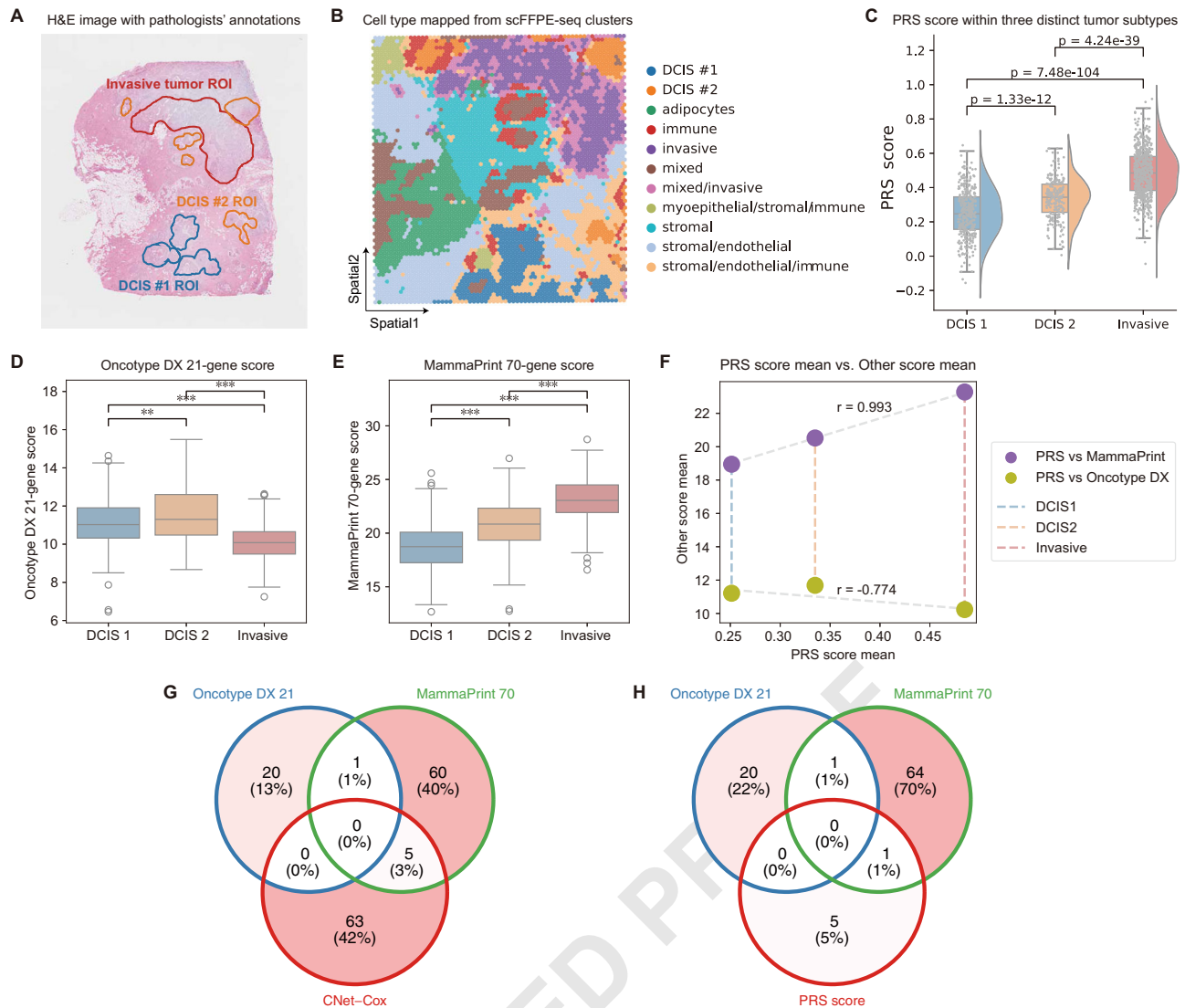


Fig. 6 | Spatial transcriptomics dataset validates the effectiveness of the PRS in prognostic analysis across distinct BRCA tumor subtypes. **A** H&E image with pathologists' annotations¹⁴. **B** Cell type mapped from scFFPE-seq clusters¹⁴. **C–E** Comparison of the PRS score, Oncotype DX⁶² 21-gene score, and MammaPrint²⁰ 70-

gene score across three distinct tumor subtypes. **F** Mean PRS scores and Pearson correlations among three distinct tumor subtypes. **G, H** Venn diagram for 68 prognostic biomarkers of the CNet-Cox model **G** and six signature genes in the PRS index **H**.

biomarkers. To address this limitation, we proposed a novel connected network-regularized Cox proportional hazards model (CNet-Cox), which enhances biological interpretability and prediction accuracy. CNet-Cox selects prognostic feature genes by integrating differential expression with prior cancer pathway information, using gene expression data, known breast cancer marker genes, and gene interaction networks from RegNetwork. This approach successfully identified 68 BRCA prognostic biomarkers, forming a connected network component that elucidates potential dysregulated pathways. Functional enrichment, semantic similarity, and anthracycline drug resistance analyses validate the significance of these biomarkers in BRCA pathways. Based on these findings, we developed a breast cancer diagnostic risk scoring index using six biomarker genes. This index effectively assesses prognostic risk and survival status, as confirmed by tumor metastasis analysis, protein expression profile analysis, and validation with external datasets, highlighting its potential for clinical application in BRCA risk monitoring systems.

Although we have primarily demonstrated the CNet-Cox framework for breast cancer prognosis, its application can be extended to identify prognostic biomarkers in other complex malignancies, including hepatocellular carcinoma, pancreatic cancer, high-grade serous ovarian cancer and

so on. However, two important considerations should be noted regarding the framework's implementation: First, CNet-Cox is inherently goal-dependent. The required prior knowledge (e.g., disease-specific marker genes) varies across different malignancies. Furthermore, when applied to tasks beyond biomarker discovery, the relevant gene interaction networks (e.g., protein-protein interactions (PPI) or ligand-receptor pairs (LR pairs)) must be carefully selected, necessitating the extraction of database-specific maximum connected component. Second, the framework exhibits data-dependence. When applied to different discovery datasets (e.g., TCGA, GEO, or METABRIC), CNet-Cox may yield distinct biomarker lists and six-gene PRS index. This variability is expected and reflects inherent differences in statistical significance across heterogeneous datasets.

In this study, CNet Cox integrates structured biological prior knowledge with connectivity constraints, whereas the baseline Cox models (Ridge-Cox, ENet-Cox, Lasso-Cox, $L_{1/2}$ -Cox, L_0 -Cox, SCAD-Cox, MCP-Cox; Supplementary Table 15) only use the corresponding gene expression features without network connectivity. Therefore, the current comparison does not fully determine whether the performance improvements of CNet-Cox are due to the proposed modeling framework or to the availability of richer prior information. However, the comparative study offers direct evidence

that the proposed CNet-Cox model, incorporating connected network constraints, transcends the independence assumption between gene variables, thereby enhancing result interpretability and improving performance in survival metrics. Currently, we only utilize prior knowledge from GRNs. In the future, we can integrate a wider array of other knowledge sources to identify more reliable biomarkers. In summary, the proposed CNet-Cox is highly adaptable and can be extended to identify prognostic biomarkers for a wide range of other complex diseases. However, there are limitations in our current work. We applied the CNet-Cox model solely to gene expression profiling data, whereas integrating multi-omics data, such as DNA methylation, copy number variations, protein complexes, metabolites, and histological H&E images, could enhance biomarker discovery. Additionally, for the 68 identified prognostic biomarker genes, we relied on literature validation and computational techniques for preliminary verification. Future research should include in vitro experiments with breast cancer patient samples to facilitate translational research and clinical applications. These steps will be crucial for advancing the utility of CNet-Cox in comprehensive biomarker discovery and personalized medicine.

Methods

Datasets and preprocessing

TCGA discovery dataset. The gene expression data for BRCA analysis were collected from the Cancer Genome Atlas (TCGA) and the Gene Expression Omnibus (GEO) databases, respectively. From the TCGA database, we retrieved data on 1093 patients, comprising individual expression profiles for 20,501 genes using `curatedTCGAData` (version 1.12.1) and `TCGAutils` (version 1.10.1), where the gene expression counts were normalized by applying a log transformation ($\log(x + 1)$). Post data preprocessing, which included the exclusion of samples with zero or negative survival times, 1080 samples with 20,220 genes remained for analysis using `colData(brca.primary.solid.tumor)`.

GEO validation dataset. For external validation, microarray datasets GSE2034, GSE7390, GSE17705, GSE20711, GSE37181 and GSE42568 (a total of 1829 patients) were sourced from GEO. These datasets, detailed in Supplementary Table 8, provided additional clinical characteristics such as patient age, tumor size, pathologic stage (I, II, III, IV, V and NA), pathology stage (T, N, M) and survival status. We utilized these datasets to identify and validate prognostic biomarkers for BRCA.

Pathological annotations. Traditionally, pathological annotations have been regarded as the “gold standard” for assessing benign and malignant tumors in cancer diagnosis and prognosis^{63,64}. Pathologists must analyze tumor tissue to accurately determine whether a tumor is benign or malignant and to establish its stage. Currently, there are various methods for tumor staging, with the T, N, M staging system being the internationally recognized standard⁶⁵, which is also the standard method used for the clinical staging of malignant tumors¹⁴. Supplementary Table 9 provides the clinical information of patients with breast cancer tumors, as documented in the American Joint Committee on Cancer (AJCC) staging manual (eighth edition)⁶⁶. It includes explanations for each parameter in the pathological stage¹ (Pathological stage) and pathological stage² (Pathology stage).

10x Visium Spatial Transcriptomic validation dataset. For the spatial transcriptomic (ST) dataset from 10x Visium¹⁴, we treated each spot as an individual sample (total 4992 spots with 18,085 genes), and processed the ST spot-by-gene matrix with `scanpy` (version 1.9.8) in Python. Genes detected in fewer than 10 spots were removed, and spot counts were normalized to the median library size using `sc.pp.normalize_total()` function. In particular, based on histopathological annotations, we then extracted spatial gene expression for three ROIs separately: DCIS 1 (344 spots), DCIS 2 (210 spots), and Invasive (599 spots).

Subtype-annotated GSE96058 cohort validation dataset. For external validation in a cohort with annotated breast cancer subtypes (ER,

HER2 and PR status), we used the GSE96058 dataset⁵⁸, a prospective population-based cohort of 3273 breast cancer cases (including 136 technical replicates) with a median follow-up of 52 months. For our analyses, we selected samples profiled on the GPL18573 platform and extracted clinical information to define ER-positive (289 patients) and TNBC subgroups (24 patients). Namely, the preprocessing of the subtype-annotated GSE96058 cohort as follows:

- Gene filtering and deduplication: To reduce noise and retain informative genes, the interquartile range (IQR) was calculated for each gene across all samples (excluding the gene identifier column). Genes were ranked by IQR in descending order. For duplicated identifiers, only the entry with the highest IQR was retained for subsequent analyses.
- Sample selection and clinical data extraction: 340 samples profiled on the GPL18573 platform are selected, and relevant clinical data are extracted, including sample ID, overall survival event, and overall survival days.
- Data normalization and subtype selection: The gene expression count matrix was standardized using z-score normalization (mean-centering and scaling to unit variance) via the R function `scale()`. Based on clinical annotations, 289 ER-positive samples (`er.status=='er status:1'`) and 24 TNBC samples (`er.status=='er status:0'`, `her2.status=='her2 status:0'`, and `pr.status=='pr status:0'`) were selected for downstream analyses.

Prior knowledge and network information integration

Initially, we utilized the `curatedTCGAData` (version 1.12.1) and `TCGAutils` (version 1.10.1) packages to acquire RNA-seq data and corresponding clinical information for BRCA samples from the TCGA database. Subsequently, we used the `DESeq2`⁶⁷ (version 1.30.1) package to analyze the gene expression data from 1080 samples, comprising 928 “surviving” and 152 “deceased” samples at the last follow-up that demonstrated by a binary indicator variable $\delta_i \in \{0, 1\}$ (Fig. 1B). Based on the criteria of $P_{\text{adj}} < 0.01$ and $|\log(\text{FC})| > 2.00$, we identified 196 significant DEGs. Recognizing the importance of previously identified genes associated with BRCA in pinpointing potential prognostic biomarkers, we integrated five categories of prior genes. These included 119 TF genes from RegNetwork³² (Supplementary Table 10), 147 genes from the breast cancer pathway in the KEGG database²¹ (Supplementary Table 11), 70 breast cancer marker genes from MammaPrint²⁰ (Supplementary Table 12), 128 breast cancer diagnostic marker genes from OSbrca³⁰ (Supplementary Table 13), and 10 breast cancer prognostic marker genes from scPrognosis³¹ (Supplementary Table 14). Subsequently, all 616 genes were mapped to the human gene regulatory network documented in RegNetwork³², and the largest connected component of this network was extracted. Ultimately, 544 genes with interconnected network relationships were identified, and potential prognostic biomarkers were determined using the CNet-Cox method we developed.

Connected network-regularized Cox proportional hazards (CNet-Cox) model

Let T represent the potential survival time for each patient. For each patient, we observe $Y = \min(T, C)$, where C is the censoring time⁶⁸. To investigate the relationship between the survival time T of patients and the corresponding gene expression levels $X_1, X_2, \dots, X_n$, we assume that the dataset $\mathcal{D}$ containing n samples is structured as follows:

$$\mathcal{D} = \{(X_1, y_1, \delta_1), (X_2, y_2, \delta_2), \dots, (X_n, y_n, \delta_n)\}, \quad (2)$$

where $X_i = (x_{i1}, x_{i2}, \dots, x_{ip})^T \in \mathbb{R}^p$ is the vector with p features of the i -th sample, $y_i \in \{-1, +1\}$ is the corresponding binary label for X_i with $i = 1, 2, \dots, n$. The key elements in the survival model are the inclusion of an

additional indicator variable δ_i , expressed as follows:

$$\delta_i = \begin{cases} 0, & \text{if } y_i \text{ is a right censoring time,} \\ 1, & \text{if } y_i \text{ is not censored.} \end{cases} \quad (3)$$

Equation (3) defines a binary indicator variable, which signifies whether $y_i (i = 1, 2, \dots, n)$ represents a censoring time or the time of death. To analyze the risk of disease recurrence or death at a given time t , David Cox proposed the influential Cox proportional hazards (CPH) model²⁸, which is expressed as follows:

$$\begin{aligned} h(t, \mathbf{X}_i) &= h_0(t) \cdot \exp(\theta_1 x_{i1} + \theta_2 x_{i2} + \dots + \theta_p x_{ip}) \\ &= h_0(t) \cdot \exp(\mathbf{X}_i^\top \boldsymbol{\theta}), \end{aligned} \quad (4)$$

where $h(t, \mathbf{X}_i)$ denotes the hazard function for the i -th sample at time t , $h_0(t)$ represents the baseline hazard (the risk when $\mathbf{X}_i = 0$), and $\boldsymbol{\theta} = (\theta_1, \theta_2, \dots, \theta_p)^\top \in \mathbb{R}^p$ is the p -dimensional vector of regression coefficients to be estimated for the i -th sample⁶⁹.

Assuming there is no correlation between samples, i.e., survival times are independent, simple calculations on Equation (4) yield the following expression for the partial likelihood function:

$$L(\boldsymbol{\theta}|\mathcal{D}) = \prod_{\{i|\delta_i=1\}} \frac{\exp(\mathbf{X}_i^\top \boldsymbol{\theta})}{\sum_{j \in H_i} \exp(\mathbf{X}_j^\top \boldsymbol{\theta})}, \quad (5)$$

where t_i denotes the survival time (either observation time or censoring time) for the i -th sample³³, and $H_i = \{j | t_j > t_i, j = 1, 2, \dots, n\}$ denotes the risk set of surviving samples at time t_i ^{70,71}.

To estimate the regression coefficient $\boldsymbol{\theta}$ using Equation (5), it is essential to maximize the log partial likelihood function, which is given by:

$$\begin{aligned} \mathcal{L}(\boldsymbol{\theta}|\mathcal{D}) &= \log L(\boldsymbol{\theta}|\mathcal{D}) \\ &= \sum_{\{i|\delta_i=1\}} \log \left[\frac{\exp(\mathbf{X}_i^\top \boldsymbol{\theta})}{\sum_{j \in H_i} \exp(\mathbf{X}_j^\top \boldsymbol{\theta})} \right] \\ &= \sum_{i=1}^n \delta_i \left\{ \mathbf{X}_i^\top \boldsymbol{\theta} - \log \left[\sum_{j \in H_i} \exp(\mathbf{X}_j^\top \boldsymbol{\theta}) \right] \right\}. \end{aligned} \quad (6)$$

Equation (6) corresponds to the conditional probability of observing a sample's death within the risk set H_i . This equation highlights a unique feature of the CPH model: the baseline hazard function $h_0(t)$ does not play a role in the practical estimation of regression coefficients, allowing the model to focus solely on the relative risks associated with covariates.

The detailed construction process of the regularized Cox proportional hazards model has been thoroughly discussed by Li and Liu¹¹, so it will not be reiterated. Here, we mainly focus on developing and solving the novel connected network-regularized Cox proportional hazards (CNet-Cox) model, and then solving the model for selecting the optimal tuning parameters, which is crucial for the effective application of the model in practical scenarios.

Model construction. Essentially, maximizing the log-partial likelihood function in Equation (6) to solve for the coefficient vector $\boldsymbol{\theta}$ is equivalent to minimizing its negative counterpart. However, the CPH model, as discussed in Li and Liu¹¹, cannot be directly used to estimate the coefficient $\boldsymbol{\theta}$ because not all p genes contribute to predicting survival outcomes. This is especially true in high-dimensional, small sample datasets such as microarrays or RNA-seq, where the number of variables p often far exceeds the number of samples n ($p > n$)¹¹. Consequently, some components $\theta_k (k = 1, 2, \dots, p)$ of the coefficient vector $\boldsymbol{\theta}$ are effectively zero. To address this, Tikhonov proposed a regularization method⁷² to achieve sparsity in solving Equation (6).

Building on the concept of "regularization", we introduce a connected network-regularized Cox proportional hazards (CNet) penalty term,

denoted as $\sum_{l \in S} \delta_l(U) \geq \delta_k(U) + \delta_j(U) - 1$ with inequality constraints $\forall S \in \Gamma(k, j), \forall k, j \in U, e_{kj} \notin E, \forall U \subseteq C \subseteq G$. This approach, previously proposed and utilized in our other work for diagnostic biomarker discovery using the CNet-SVM model²⁵, is applied to the negative formula of Equation (6). By incorporating network connectivity inequality constraints⁷³ and minimizing the objective function, we obtain the CNet-Cox model, which effectively determines the coefficient vector $\boldsymbol{\theta}$:

$$\begin{aligned} \text{Minimize} \quad & -\mathcal{L}(\boldsymbol{\theta}; \mathcal{D}) + \alpha \lambda \sum_{j=1}^p |\theta_j| + (1 - \alpha) \frac{\lambda}{2} \boldsymbol{\theta}^\top \mathbf{L} \boldsymbol{\theta} \\ & \sum_{l \in S} \delta_l(U) \geq \delta_k(U) + \delta_j(U) - 1, \\ \text{Subject to} \quad & \forall S \in \Gamma(k, j), \forall k, j \in U, e_{kj} \notin E, \forall U \subseteq C \subseteq G. \end{aligned} \quad (7)$$

where $\mathcal{L}(\boldsymbol{\theta}; \mathcal{D})$ is defined by Equation (6), $\mathbf{L}$ denotes the normalized graph Laplacian matrix⁷³, the parameters $\lambda > 0$ and $\alpha \in (0, 1)$ are tuning parameters used to balance the loss and penalty functions within the model, which play a crucial role in adjusting the model to achieve optimal performance by controlling the trade-off between fitting the data accurately and imposing regularization for sparsity and connectivity.

Model solving

The partial log-likelihood function $\mathcal{L}(\boldsymbol{\theta}; \mathcal{D})$, as defined by Equation (6), is expressed as $\sum_{i=1}^n \delta_i \left\{ \mathbf{X}_i^\top \boldsymbol{\theta} - \log \left[\sum_{j \in H_i} \exp(\mathbf{X}_j^\top \boldsymbol{\theta}) \right] \right\}$. It is evident that this function is essentially a summation term conditioned on "survival time" and "censoring status" with "survival time" being nested within "censoring status". While it is feasible to use nested loops in programming to compute and solve this function, the "loop" operation tends to be inefficient and slow during program execution. Thus, we aim to expedite the solution process by employing matrices. This approach necessitates transforming the general form of the partial log-likelihood in Equation (6) into an equivalent mathematical matrix representation, thereby enhancing computational efficiency and suitability for large-scale data analysis.

To begin with, define the indicative matrix $H \in \mathbb{R}^{n \times n}$ with elements h_{ij} :

$$h_{ij} = \begin{cases} 1, & t_i < t_j, \\ 0, & \text{others.} \end{cases} \quad (8)$$

This matrix will be used to replace the risk set H_i , which reflects "survival time" in Equation (6). Next, define the characteristic vector $\boldsymbol{\delta}$ as follows:

$$\boldsymbol{\delta} = (\delta_1, \delta_2, \dots, \delta_n)^\top \in \mathbb{R}^{n \times 1}. \quad (9)$$

Obviously, Equation (9) holds $\boldsymbol{\delta}(i) = 1$. We will use this vector to replace the conditional summation operation $\sum_{\{i|\delta_i=1\}}$ in Equation (6), which accounts for the "truncation censoring state". Finally, based on Equations (8) and (9), the partial log-likelihood function $\mathcal{L}(\boldsymbol{\theta}; \mathcal{D})$ shown in Equation (6) can be rewritten as follows:

$$\begin{aligned} \mathcal{L}(\boldsymbol{\theta}|\mathcal{D}) &= \sum_{i=1}^n \delta_i \left\{ \mathbf{X}_i^\top \boldsymbol{\theta} - \log \left[\sum_{j \in H_i} \exp(\mathbf{X}_j^\top \boldsymbol{\theta}) \right] \right\} \\ &= \boldsymbol{\delta}^\top \cdot \left\{ \mathbf{X} \boldsymbol{\theta} - \log \left[(\exp(\mathbf{X}_1^\top \boldsymbol{\theta}), \exp(\mathbf{X}_2^\top \boldsymbol{\theta}), \dots, \exp(\mathbf{X}_n^\top \boldsymbol{\theta})) \odot_{\text{row}}^{\text{sum}} \mathbf{H} \right]^\top \right\}, \end{aligned} \quad (10)$$

where $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2, \dots, \mathbf{X}_n]^\top \in \mathbb{R}^{n \times p}$.

Here, the Hadamard product $\odot_{\text{row}}^{\text{sum}}$ is called the term-by-term (row-wise) product. Let $\mathbf{h} = (\exp(\mathbf{X}_1^\top \boldsymbol{\theta}), \exp(\mathbf{X}_2^\top \boldsymbol{\theta}), \dots, \exp(\mathbf{X}_n^\top \boldsymbol{\theta}))$, then $\mathbf{h} \odot_{\text{row}}^{\text{sum}} \mathbf{H}$ means that the Hadamard product operation is first applied between the vector $\mathbf{h}$ and each row vector of the matrix $\mathbf{H}$. Following this, the resulting vectors are then summarized (e.g., by summing their components).

More generally, the Hadamard product is a binary operation between two matrices (or vectors) of the same dimensions, producing an output matrix (or vector) of the same size. Each element of the output is the product of the corresponding elements from the two input matrices (or vectors). For illustration, when the sample size $n = 3$, the operation proceeds as follows:

Let $\mathbf{h} = (h_1, h_2, h_3)$ be a vector and $H = (h_{ij}) \in \mathbb{R}^{3 \times 3}$ be a matrix. Then we have

$$\begin{aligned} \mathbf{h} \odot_{\text{row}}^{\text{sum}} H &= (h_1, h_2, h_3) \odot_{\text{row}}^{\text{sum}} \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \\ &= \left(\sum_{i=1}^3 h_i \cdot h_{i1}, \sum_{i=1}^3 h_i \cdot h_{i2}, \sum_{i=1}^3 h_i \cdot h_{i3} \right). \end{aligned} \quad (11)$$

However, since MATLAB R2020a does not provide a direct implementation of the $\odot_{\text{row}}^{\text{sum}}$ operation, we performed an equivalent transformation of Equation (11) as follows:

$$\mathbf{h} \odot_{\text{row}}^{\text{sum}} H = (h_1, h_2, h_3) \cdot \begin{bmatrix} h_{11} & h_{21} & h_{31} \\ h_{12} & h_{22} & h_{32} \\ h_{13} & h_{23} & h_{33} \end{bmatrix} = \mathbf{h} \cdot H^T. \quad (12)$$

Based on Equations (11) and (12), then Equation (10) can be equivalently rewritten as:

$$\mathcal{L}(\boldsymbol{\theta}|\mathcal{D}) = \delta^T \cdot \left\{ \mathbf{X}\boldsymbol{\theta} - \log \left[\exp(\mathbf{X}_1^T \boldsymbol{\theta}), \exp(\mathbf{X}_2^T \boldsymbol{\theta}), \dots, \exp(\mathbf{X}_n^T \boldsymbol{\theta}) \right] \cdot H^T \right\}^T. \quad (13)$$

The CNet-Cox model (7) was initially a constrained optimization problem under a given risk set. However, Equations (10) and (13) remove the risk set's summation condition from the objective function, thereby converting the model into a conventional/standard constrained optimization problem. From this, it can be seen that the original problem (7) can be solved by applying the Sequential Unconstrained Minimization (SUMT) interior point method, by shrinking some components of the regression coefficient $\boldsymbol{\theta}$ to zero, thus realizing the function of feature selection.

Optimal parameter selection. Without loss of generality and consistent with previous discussion, we set the parameter α to 0.5. For the CNet-Cox model, the tuning parameter λ is determined using K -fold cross-validation (CV)⁷⁴. The discovery dataset is divided into K -folds, with $(K - 1)$ -folds used for training and the remaining fold for validation. This process is repeated K times, allowing the test data to select parameters that provide the best fit to the training data and the best performance on new data.

The optimal λ value is obtained by minimizing the cross-validation log partial likelihood (CV-LPL)³³, which is the sum of the individual sample log partial likelihoods. This approach has demonstrated strong performance in Cox survival analysis models⁷⁵. The CV-LPL is defined as:

$$\text{CV} - \text{LPL}(\lambda) = - \sum_{k=1}^K \left\{ \mathcal{L}(\hat{\boldsymbol{\theta}}^{(-k)}(\lambda)) - \mathcal{L}^{(-k)}(\hat{\boldsymbol{\theta}}^{(-k)}(\lambda)) \right\}, \quad (14)$$

where $\hat{\boldsymbol{\theta}}^{(-k)}(\cdot)$ represents the estimate of $\boldsymbol{\theta}$, which is obtained when the k th fold data is excluded from the given λ of the model. $\mathcal{L}(\cdot)$ is the log partial likelihood using all n samples, and $\mathcal{L}^{(-k)}(\cdot)$ is the log partial likelihood excluding the k th fold sample²³. It is worth noting that the choice of K usually depends on the size of the dataset. When K is given, the coefficient $\hat{\boldsymbol{\theta}}^{(-k)}(\lambda)$ ⁶⁸ can be calculated accordingly.

Feature selection via regression coefficient threshold. Feature selection in CNet-Cox model is governed by a threshold parameter, τ , applied to the regression coefficients θ_j . Specifically, a feature is retained if its corresponding coefficient satisfies $|\theta_j|/\|\boldsymbol{\theta}\|_{\infty} \geq \tau$, where $\|\boldsymbol{\theta}\|_{\infty}$ represents the largest absolute coefficient value. Coefficients that do not meet this threshold are set to zero, and their associated features are excluded from the model. This strategy follows the cardinality constraint approach described by Chan et al.³⁴, where feature selection is based on the relative magnitude of model coefficients. In this work, to determine an appropriate value for τ , we systematically assessed CNet-Cox performance on the discovery cohort across a range of candidate τ values. Model performance remained stable for τ values between $5.00\text{e-}2$ and $5.50\text{e-}2$. The final value of τ was determined solely using the discovery cohort, without reference to the test set or external validation cohorts, thereby reducing the risk of overfitting.

Network topology analysis

Figure 2C shows the degree distribution of nodes in the feature network, which adheres to a power-law distribution, indicating that the network exhibits a scale-free topology. This is consistent with the widely observed phenomenon in real biomolecular networks, where the majority of connections are concentrated among a small number of highly connected hub nodes. In graph theory, betweenness centrality quantifies the importance of a node based on the number of shortest paths that pass through it. Nodes with high betweenness centrality are critical for maintaining network connectivity, as their removal can significantly disrupt network connectivity. Figure 2D illustrates the relationship between betweenness centrality and node degree in the feature network, with a correlation coefficient of $R = 0.8466$, indicating a strong positive association. Such a relationship between hubness and centrality is commonly observed in real-world biological networks, where a small number of central regulators mediate communication between many peripheral nodes. Together, Fig. 2C, D (generated using Cytoscape⁷⁶, version 3.10.3) indicate that the prognostic network identified by CNet-Cox recapitulates key topological properties of biomolecular networks, supporting the biological plausibility and non-random organization of the selected biomarkers.

Functional enrichment analysis with breast cancer

For the 68 prognostic biomarker genes obtained, Metascape (<https://metascape.org>) performs pathway and biological process enrichment analyses using the following seven ontology sources: KEGG Pathway, GO Biological Processes, Reactome Gene Sets, Canonical Pathways, CORUM, WikiPathways, and PANTHER Pathway. All genes in the genome are used as the enrichment background set³⁶. Metascape filters terms with P -value < 0.01 , a counts > 3 , and enrichment factors > 1.5 , grouping them based on their member similarity. The P values are computed using the cumulative hypergeometric distribution. During hierarchical clustering of enriched terms, the Kappa score is utilized as a measure of similarity, with subtrees having a Kappa score > 0.3 considered as a class. The most statistically significant term within each cluster is ultimately selected to represent the cluster.

To investigate the relevance of prognostic biomarkers to breast cancer signaling, we analyzed the top five enriched pathways identified among the 68 selected biomarkers. As shown in Supplementary Fig. 4A, these pathways are closely associated with breast cancer, with the top five distinctly highlighted. Nodes are color-coded by GO term clusters, indicating functional grouping. Notably, previous studies have directly linked these pathways to breast cancer development and progression. For example, the second-ranked pathway ("WP5087", $P = 1.0\text{e-}31$; Fig. 3A) from WikiPathways was implicated in malignant pleural mesothelioma following breast cancer radiotherapy³⁷. The third-ranked pathway ("R-HSA-8939211", $P = 1.0\text{e-}23$) from Reactome involves ESR-mediated signaling and distant metastasis of Luminal A breast cancer³⁸. The fourth ("hsa05166", $P = 1.0\text{e-}18$) from KEGG highlights the role of Human T-cell leukemia virus type 1 (HTLV-1) in sensitizing breast cancer cells to malignant transformation^{39,77}, and the

fifth ("M167", $P \approx 1.0e-17$) from Canonical Pathways pertains to PID-API regulation of breast cancer initiation⁴⁰.

To further characterize the biological functions of these biomarkers, we performed PPI enrichment analysis using STRING, BioGrid, OmniPath, and InWeb_IM databases. Based on STRING (physical score > 0.132) and BioGrid data, a PPI network was constructed containing 63 genes and 255 edges among the 68 biomarkers (Supplementary Fig. 4B). The Molecular Complex Detection (MCODE) algorithm was applied to detect network modules, resulting in four connected components (Supplementary Fig. 4C). Pathway enrichment analysis of these modules revealed terms consistent with those identified in the overall network, as shown in Supplementary Fig. 4D, further supporting the biological relevance of the connected components represented by red nodes in Supplementary Fig. 4C.

Semantic similarity measure with unique GO terms of breast cancer

GO is a gene ontology term set aimed at consistent gene product descriptions across various databases. The GO annotation of gene function can be used to measure the similarity of gene function, and the quality of GO semantic similarity has been confirmed. Suppose there are two GO term sets: $\mathcal{G}_1 = \{go_{11}, go_{12}, \dots, go_{1k}\}$ and $\mathcal{G}_2 = \{go_{21}, go_{22}, \dots, go_{2l}\}$, let the similarity of each node go_{1i} ($i = 1, 2, \dots, k$) belonging to set $\mathcal{G}_1$ and set $\mathcal{G}_2$ be the maximum value of the similarity of each term go in set $\mathcal{G}_1$ and set $\mathcal{G}_2$, then it holds that

$$\text{Sim}(go_{1i}, \mathcal{G}_2) = \max_{1 \leq j \leq l} (S_{\mathcal{G}_2}(go_{1i}, go_{2j})). \quad (15)$$

Correspondingly, the similarity between each term go_{2j} ($j = 1, 2, \dots, l$) in set $\mathcal{G}_2$ and set $\mathcal{G}_1$ is

$$\text{Sim}(go_{2j}, \mathcal{G}_1) = \max_{1 \leq i \leq k} (S_{\mathcal{G}_1}(go_{2j}, go_{1i})). \quad (16)$$

Therefore, the semantic similarity measure (SS-measure) of $\mathcal{G}_1$ and $\mathcal{G}_2$ is defined as the arithmetic average of the above two similarity indicators, which is

$$\text{SS-measure}(\mathcal{G}_1, \mathcal{G}_2) = \frac{\max_{1 \leq i \leq k} \text{Sim}(go_{1i}, \mathcal{G}_2) + \max_{1 \leq j \leq l} \text{Sim}(go_{2j}, \mathcal{G}_1)}{k + l}, \quad (17)$$

where $S_{\mathcal{G}}$ is defined by Equation (3) in Wang et al.'s work⁷⁸.

Pathological staging in clinical practice

Nomograms provide clinicians with valuable tools for individualized treatment planning, aiming to enhance both survival outcomes and quality of life for breast cancer patients. To evaluate the prognostic significance of the PRS metric in clinical stratification, we classified patient data using key clinicopathological variables (Supplementary Table 16): gender (male = 0, female = 1), pathologic stage (stage I = 0; stages II–V = 1), and TNM stages (T1/T2/Tx = 0, T3/T4 = 1; N0/N1 variants = 0, N1/N2/N3 variants = 1; M0/cM0/Mx = 0, M1 = 1). These groupings enabled systematic assessment of clinical features in relation to patient prognosis, visualized through nomograms and calibration curves. Using the discovery dataset, we constructed a nomogram in which the length of each line segment reflects the relative contribution of each variable to patient outcomes. Individual scores are summed to generate a total score, which is then mapped to estimated survival probabilities. Model accuracy was further evaluated using calibration curves, where the x -axis represents the survival probabilities predicted by the nomogram, while the y -axis represents the actual survival probabilities. The 45-degree diagonal line shown in blue represents the benchmark for ideal predictions. The closer the red fitted line is to the blue diagonal line, the higher the accuracy of the nomogram¹⁵.

Survival information in clinical

It's noteworthy that Fig. 5A includes more than one type of survival information, as breast cancer patients often have multiple clinical trial endpoints from diagnosis to remission or death. Overall Survival (OS) in Fig. 5A refers to the time from diagnosis or treatment initiation to death from any cause. This comprehensive survival metric evaluates the overall impact of the disease on a patient's life. Disease-Free Survival (DFS) in Fig. 5A refers to the time from diagnosis or treatment initiation to disease recurrence (local, regional, or distant) or death from any cause. This metric assesses the effectiveness of treatment by measuring how long a patient can maintain a disease-free state post-treatment. Recurrence-Free Survival (RFS) in Fig. 5A refers to the time from diagnosis or treatment initiation to disease recurrence (local, regional, or distant). This metric focuses on how long a patient can avoid disease recurrence after treatment, aiding in understanding the long-term effectiveness of treatment strategies. Distant Metastasis-Free Survival (DMFS) in Fig. 5A refers to the time from diagnosis or treatment initiation to the occurrence of distant metastasis. This metric evaluates how long a patient can avoid distant metastasis post-treatment, which is often associated with poor prognosis. Overall, survival analysis was conducted on 1,829 breast cancer patients from eight independent external validation datasets. The results indicate that the PRS metric established in this paper has a significant impact on breast cancer prognosis.

External validation of PRS in independent datasets

For all external validation analyses using independent GEO (microarray or RNA-seq) or Visium ST RNA-seq datasets, no DEG selection was performed on these external cohorts. Instead, the six prognostic biomarker genes (included in the PRS metric) identified from the TCGA discovery dataset were directly validated in the external datasets. Specifically, we used the expression profiles of these prognostic biomarker genes in the independent datasets and applied the PRS score (i.e., Equation (1)) from the Cox model trained on the TCGA dataset to assess their prognostic value. Our aim at this stage was not to re-identify DEGs but to rigorously evaluate whether the combination of these six prognostic biomarkers could jointly predict patient outcomes in new cohorts. This approach ensures that external validation is entirely independent and free from information leakage, as no information from the external datasets was used during biomarker discovery or model training.

Comparing PRS score with MammaPrint and Oncotype DX in spatial transcriptomics data

Within three ROIs (DCIS 1: 344 spots; DCIS 2: 210 spots; Invasive: 599 spots) shown in Fig. 5C, we computed PRS, MammaPrint, and Oncotype DX scores as the mean of $\log(\text{normalized counts} + 1)$ across genes in each signature (see PRS gene, Supplementary Table 8, and Supplementary Table 12). The resulting signature scores were stored in `ada.obs` in the `AnnData` object. We visualized their distributions across ROIs with boxplots (`matplotlib` version 3.7.5, `seaborn` version 0.13.2) and assessed between-ROI statistical significance using `scipy` (version 1.10.1) in Python. Next, we compare Pearson correlations between PRS and MammaPrint or Oncotype DX across all selected spots (PRS vs. MammaPrint: $r = 0.442$; PRS vs. Oncotype DX: $r = -0.212$, respectively). To account for tissue heterogeneity across distinct cell types, we averaged the three scores within each ROI and found the PRS means were 0.251 (DCIS 1), 0.334 (DCIS 2), and 0.485 (Invasive). Treating these ROI-level means as a single vector summarizing tissue heterogeneity, the Pearson correlations of score mean between the PRS and MammaPrint or Oncotype DX were $r = 0.993$ and $r = -0.773$, respectively.

Data availability

The biomarker discovery dataset of breast cancer was downloaded from the TCGA <https://cancergenome.nih.gov/>. The independent validation datasets of breast cancer were downloaded from the GEO under accession numbers [GSE42568](#), [GSE20711](#), [GSE37181](#), [GSE17705](#), [GSE2034](#), [GSE7390](#). Another independent validation dataset of estrogen receptor-positive (ER-positive)

breast cancer and triple-negative breast cancer (TNBC) patients was downloaded from the GEO under accession number [GSE96058](#) with GPL18573 platform (last modified: 2025-11-01, 01:03). The raw and processed 10x Visium spatial sequencing data of breast cancer tissues were downloaded from 10x Genomics <https://www.10xgenomics.com/products/xenium-in-situ/preview-dataset-human-breast>, and GEO under accession number [GSE243280](#). The external validation data for breast cancer anthracycline resistance analysis can be downloaded from the TCGA <https://cancergenome.nih.gov/>, and was directly accessed from Supplementary Table S2 with “Differentially expressed genes between the anthracycline-S and anthracycline-R patients” from Liu et al.’s work⁴⁴. The external validation dataset of breast cancer metastasis analysis was downloaded from GEO under accession numbers [GSE147995](#).

Code availability

All R, MATLAB, and Python scripts, along with the input and output data used in this study, are available in the GitHub repository at <https://github.com/zpliulab/CNet-Cox>. Other large input datasets can be found at Google Drive.

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Author contributions

Z.L. conceived and administered the CNet-Cox study. L.L. designed the analytical approach, performed model development, baseline implementation, data analysis and curation, and wrote the software. W.Z. and Q.Z. provided expert advice, reviewed the manuscript formatting, and critically revised the manuscript. L.L., W.C. and Z.L. wrote, revised and contributed to the final manuscript. All authors have critically read and approved the final manuscript.

Competing interests

Qingpeng Zhang is an associate editor of npj Digital Medicine. The concerned author is not part of the peer review process or decision-making of the manuscript.

Additional information

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Correspondence and requests for materials should be addressed to Wai-Ki Ching or Zhi-Ping Liu.

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